Human influences on ecosystems

30 questions: 15 multiple choice, 15 structured.

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1 ? Part 1 Core Food supply [1]

How can a herbicide increase crop yield?

A by supplying every mineral ion directly

B by reducing competition between weeds and crop plants

C by increasing insect damage

D by stopping all photosynthesis in the crop

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2 ? Part 1 Core Food supply [1]

What is a disadvantage of a large monoculture?

A all crops must have different harvesting times

B machinery cannot be used

C the field necessarily contains more plant species than a mixed crop

D a disease adapted to the crop may spread through many susceptible plants

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3 ? Part 1 Core Food supply [1]

Which is a potential disadvantage of intensive livestock production?

A rapid spread of disease among closely housed animals

B the complete absence of animal waste

C no need to supply food

D guaranteed elimination of all welfare concerns

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4 ? Part 1 Core Habitat destruction [1]

Which measurement is the syllabus definition of biodiversity in an area?

A the mass of the largest animal

B the total number of individuals of one species

C the number of different species

D the amount of rainfall alone

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5 ? Part 1 Core Habitat destruction [1]

Which activity directly destroys a terrestrial habitat?

A counting birds without disturbing them

B storing seeds in a conservation bank

C designating an existing forest as protected

D clearing woodland to build houses

6 ? Part 1 Core Habitat destruction [1]

Why can deforestation increase soil loss after rain?

A There are always more roots after clearing.

B Roots no longer bind the soil effectively.

C All rainwater is immediately converted to wood.

D The soil becomes impossible to move.

7 ? Part 1 Core Pollution [1]

Untreated sewage enters a river and fish die after dissolved oxygen falls. Which process contributes most directly to oxygen loss?

A photosynthesis by every fish

B oxygen production by bacteria in all conditions

C aerobic respiration by decomposers

D conversion of oxygen into sunlight

8 ? Part 1 Core Pollution [1]

Why can non-biodegradable plastic remain a long-term environmental problem?

A It is not readily broken down by decomposers.

B It always dissolves immediately in water.

C It is always used as food safely by animals.

D It prevents any animal becoming entangled.

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9 ? Part 1 Core Pollution [1]

Which pair of gases contributes to the enhanced greenhouse effect?

A oxygen and nitrogen only

B methane and carbon dioxide

C helium and neon only

D hydrogen and argon only

10 ? Part 1 Core Conservation [1]

Which resource use is sustainable?

A removing a population faster than it reproduces indefinitely

B clearing all forest without replacement

C harvesting at a rate no greater than replenishment over time

D catching every breeding fish each season

11 ? Part 1 Core Conservation [1]

What is the main purpose of a seed bank?

A ensuring all seeds are eaten immediately

B removing every plant from protected habitats

C replacing all conservation with one crop variety

D preserving viable seeds as a reserve of plant genetic material

12 ? Part 1 Supplement Conservation [1]

Why can a larger fishing-net mesh help conserve fish stocks?

A It allows more small immature fish to escape.

B It catches every young fish before breeding.

C It removes all predators from the sea.

D It prevents any adult fish reproducing.

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13 ? Part 1 Supplement Conservation [1]

Which measure most directly replaces trees removed for timber?

A increasing unregulated logging

B removing seedlings every year

C replanting suitable trees

D converting every cleared area to roads

14 ? Part 1 Supplement Conservation [1]

How can artificial insemination aid a captive-breeding programme?

A Sperm can be transferred between selected animals without transporting a mating pair.

B It produces offspring without any paternal genes.

C It replaces fertilisation with seed germination.

D It guarantees that every offspring is a clone.

15 ? Part 1 Supplement Conservation [1]

Why can loss of genetic variation increase extinction risk?

A Every remaining individual becomes better adapted to all conditions.

B Fewer individuals may possess alleles suited to a new environmental challenge.

C The species no longer needs a habitat.

D Disease can no longer spread.

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16 ? Part 1 Core Food supply [4]

A farmer uses machinery, fertiliser and insecticide to increase production.

(a) Describe how machinery and fertiliser can improve yield. [2]

(b) Describe how insecticide and herbicide can improve crop production. [2]

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17 ? Part 1

Core

Food supply

[4]

One farm grows a single crop over a large area; another keeps many livestock in a small indoor space.

(a) State one advantage and one disadvantage of the monoculture. [2]

(b) State one advantage and one disadvantage of intensive livestock production. [2]

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18 ? Part 1

Core

Habitat destruction

[4]

A coastal habitat is removed for construction, eliminating the main breeding area of a small fish species.

(a) Explain two possible consequences for the fish population. [2]

(b) Explain how a decline could affect the food web. [2]

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19 ? Part 1

Core

Habitat destruction

[4]

A wooded hillside is cleared before a season of heavy rain.

(a) Explain why the clearing can increase soil erosion and flooding. [2]

(b) Explain why deforestation can increase atmospheric carbon dioxide. [2]

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20 ? Part 1

Core

Habitat destruction

[4]

A forest initially contains 48 recorded species. After habitat clearance, only 18 remain in the survey area.

(a) Calculate the decrease in recorded species and state what has happened to biodiversity. [2]

(b) Explain why local habitat destruction can cause global extinction in some cases. [2]

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21 ? Part 1 Supplement Pollution [4]

Fertiliser washes into a lake, followed by a large increase in algae and later fish deaths.

(a) Explain the initial increase in algae and the subsequent rise in decomposition. [2]

(b) Explain how this sequence can kill fish. [2]

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22 ? Part 1 Core Pollution [4]

Plastic waste enters a river and neighbouring fields.

(a) Describe two ways plastic can harm animals. [2]

(b) Explain why non-biodegradable plastic can persist and spread. [2]

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23 ? Part 1 Core Pollution [4]

A region burns more fossil fuel and expands cattle farming.

(a) Identify a greenhouse gas associated with each activity. [2]

(b) Explain how increased greenhouse-gas concentrations can affect climate. [2]

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24 ? Part 1 Core Conservation [4]

A forest supplies timber and a coastal area supplies fish.

(a) Describe what makes these resources sustainable. [2]

(b) State one management measure for each resource. [2]

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25 ? Part 1 Core Conservation [4]

An island bird declines after an introduced predator arrives.

(a) Explain why this bird may become endangered. [2]

(b) Describe two conservation actions suitable for this situation. [2]

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26 ? Part 1 Supplement Conservation [4]

A community wants timber income while retaining its forest.

(a) Explain how quotas and replanting help conserve it. [2]

(b) Explain the value of protected areas and education. [2]

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27 ? Part 1

Supplement

Conservation

[4]

Monitoring shows that a fish stock has fewer breeding adults each year.

(a) Explain how closed seasons and catch quotas can help. [2]

(b) Explain why continued monitoring and suitable mesh sizes are useful. [2]

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28 ? Part 1 Supplement Conservation [4]

A conservation programme protects a wetland rather than one commercially valuable species alone.

(a) Explain two reasons for maintaining biodiversity. [2]

(b) Explain two ecosystem benefits of protecting the whole habitat. [2]

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29 ? Part 1

Supplement

Conservation

[4]

A zoo keeps members of a rare mammal species at distant sites.

(a) Describe how artificial insemination can support breeding between them. [2]

(b) Describe how IVF differs from this method. [2]

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30 ? Part 1 Supplement Conservation [4]

Only a small number of individuals remain in an endangered species.

(a) Explain why low genetic variation can threaten future survival. [2]

(b) Explain why a breeding programme should aim to retain genetic variation. [2]

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